Benjamin Atawiah

Ph.D. Candidate in Mathematics
Numerical Analysis and Scientific Computing for PDEs

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RESEARCH INTERESTS

Numerical analysis and scientific computing for partial differential equations, with a focus on:

- Discontinuous Galerkin finite element methods
- Runge–Kutta time integration and strong stability preserving schemes
- Exponential and implicit–explicit (IMEX) time integration methods
- Stability and error analysis of high-order numerical schemes
- Fourier analysis and energy methods for space–time discretizations

EDUCATION

Ph.D. in Mathematics Expected Aug. 2026
The University of Alabama
Dissertation Advisor: Dr. Zheng Sun

M.A. in Mathematics Apr. 2025
The University of Alabama
Advisor: Dr. Zheng Sun

M.Sc. in Mathematical Sciences Jun. 2020
African Institute for Mathematical Sciences (AIMS), Ghana
Advisor: Dr. Mohamed Mbehou

B.A. in Economics and Mathematics Jun. 2018
University of Ghana
Advisor: Mr. Antwi Asare

PUBLICATIONS

Submitted

1. Atawiah, B., and Sun, Z. Stability and accuracy effects of filtering on Runge–Kutta discontinuous Galerkin methods: Fourier analysis for the second-order case. Submitted for publication, 2025.

In Preparation

2. Atawiah, B., and Sun, Z. Energy-based stability and error estimates for stage-dependent Class A Runge–Kutta discontinuous Galerkin methods. Manuscript in preparation.
3. Atawiah, B., and Sun, Z. Stability analysis of exponential time differencing with local discontinuous Galerkin methods for dispersive equations with generalized even-order spatial derivatives. Work in progress.
4. Atawiah, B., and Sun, Z. Local discontinuous Galerkin methods with compact stencils. Work in progress.

CONFERENCE PRESENTATIONS

- *Filtered RKDG Methods: Stability and Accuracy via Fourier Analysis.* SIAM Southeast Atlantic Section (SEAS) Conference, University of Tennessee, Knoxville. Oral presentation, Mar. 21–23, 2025.
- *Filtered RKDG Methods: Stability and Accuracy via Fourier Analysis.* Southeast Applied and Computational Mathematics Student Workshop. Oral presentation, Apr. 5–6, 2025.
- Third Joint Alabama–Florida Conference on Differential Equations, Dynamical Systems, and Applications, University of Alabama at Birmingham. Poster presentation, May 20–22, 2025.

Workshop Participation

- Winter School: Reduced-Order Modeling for Complex Engineering Problems, Institute for Mathematical and Statistical Innovation (IMSI), University of Chicago. Jan. 29–31, 2025. *(Full scholarship recipient)*
- Mathematical Modeling of Biological Interfacial Phenomena, IMSI, University of Chicago (online). Dec. 9–13, 2024.

RESEARCH EXPERIENCE

Graduate Research Assistant Fall 2024 – Present
Department of Mathematics, The University of Alabama
Advisor: Dr. Zheng Sun

- Investigate stability and accuracy of high-order numerical methods for PDEs using Fourier and energy-based analysis.
- Develop Fourier analysis framework for filtered RKDG methods coupled with local time marching and optimal CFL conditions.
- Derive energy-based stability and error estimates for stage-dependent RKDG methods.
- Analyze exponential time differencing methods for dispersive equations with higher-order spatial derivatives.
- Develop compact stencil formulations for local discontinuous Galerkin methods.

M.Sc. Thesis Research May 2020 – Jun. 2020
African Institute for Mathematical Sciences (AIMS), Ghana

- Conducted dynamical systems analysis of strange attractors in a climate-related mathematical model.
- Applied nonlinear dynamical systems techniques to study long-term behavior and stability of solutions.

TEACHING EXPERIENCE

Instructor of Record Aug. 2022 – Present
Department of Mathematics, The University of Alabama

- Applied Differential Equations I (Spring 2024, Spring 2025, Fall 2025, Spring 2026)
- Calculus II (Fall 2024)
- Calculus III (Fall 2023)
- Calculus I (Spring 2023)
- Algebra and Trigonometry (Fall 2022)

Mathematics Tutor Summers 2022–2024
Department of Mathematics, The University of Alabama

- Calculus I, Calculus II, and Calculus III

Teaching Assistant Aug. 2018 – Jul. 2019
University of Ghana

- Vectors and Geometry; Calculus I

HONORS, AWARDS, AND FELLOWSHIPS

- Summer Research Fellowship, Department of Mathematics, University of Alabama 2025
- Graduate Student Travel Award, Graduate School, University of Alabama 2025
- Travel Award, SIAM SEAS Conference 2025
- Travel Award, Southeast Applied and Computational Mathematics Student Workshop 2025
- Graduate Student Travel Funding, JAF DEDS Conference 2025
- Full Scholarship, IMSI Winter School on Reduced Order Modeling 2025
- Edith & Richard Ainsworth Endowed Scholarship, University of Alabama 2022, 2023
- AIMS Ghana Fellowship ($25,000) 2019–2020

PROFESSIONAL SERVICE

- Judge, SIMODE (SCUDEM X 2025)
- Judge, SIAM MathWorks Math Modeling (M3) Challenge (2025, 2026)
- Research Judge, Undergraduate Research and Creative Activity Conference, University of Alabama
- Volunteer Grader, MathCounts Competition (2022–2024)
- Event Support, SEC Geometry–Topology Workshop, University of Alabama
- Mathematics Tutor, Korleman Educational Outreach, Ghana

TECHNICAL SKILLS

Python, MATLAB, Mathematica, Firedrake, LaTeX

PROFESSIONAL MEMBERSHIPS

- American Mathematical Society (AMS) 2022–Present

- Society for Industrial and Applied Mathematics (SIAM) 2022–Present

REFERENCES

Available upon request.